## Task 4: Statistical Inference

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### 4.1 Comparison of Means - Test A: HB by Gender

**Hypotheses**

- $H_0$: Mean HB (Female) = Mean HB (Male)
- $H_1$: Mean HB (Female) ≠ Mean HB (Male)

**Test justification:** A two-sample t-test compares mean between two independent groups. We use **Welch's t-test** (unequal-variance version) rather than the classical t-test because Task 3's descriptive statistics already show unequal group sizes (Female n=15,597, Male n=11,437) and - as confirmed formally in Test C below - HB variances differ meaningfully by subgroup in this dataset. Welch's test does not assume equal variances, making it the more defensible default here.

**Results**

| Group | n | Mean HB (g/dL) | SD |
|---|---|---|---|
| **Female** | 15,597 | 11.61 | 1.45 |
| **Male** | 11,437 | 13.26 | 1.93 |

Welch two-sample t-test: **$t = -76.82$, $df = 20{,}293$, $p < 2.2 \times 10^{-16}$**
95% CI for the difference (Female − Male): **[−1.69, −1.61] g/dL**

**Interpretation:** The difference is statistically significant at any conventional threshold, and importantly is also **clinically meaningful**, not just statistically detectable given the large sample. Males average roughly 1.65 g/dL higher hemoglobin than females, consistent with well-established physiology (menstruation, and WHO's use of sex-specific HB thresholds, which this dataset's own who_hb_threshold field already encodes).

**Organizational implications:** This result validates that the WHO's sex-differentiated thresholds (already applied in our anemia classification) are appropriate for this population - a single pooled threshold would systematically misclassify one sex. For the client, this reinforces that **any screening or resource-planning tool must remain gender-stratified**.

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### 4.2 Comparison of Means - Test B: Young Adult Women vs. Elderly Men

**Hypotheses**

- $H_0$: Mean HB (Female, 18–39) = Mean HB (Male, 80+)
- $H_1$: Mean HB (Female, 18–39) ≠ Mean HB (Male, 80+)

**Test justification:** This test directly follows up on Task 3's most striking descriptive finding - that anemia prevalence is similar in magnitude for young women and elderly men, despite very different underlying physiology. A Welch t-test again suits this comparison of two independent groups with unequal sizes/variances.

**Results**

| Group | n | Mean HB (g/dL) | SD |
|---|---|---|---|
| **Female, 18–39** | 5,783 | 11.83 | 1.36 |
| **Male, 80+** | 251 | 11.14 | 1.97 |

Welch two-sample t-test: **$t = 5.46$, $df = 260.4$, $p = 1.09 \times 10^{-7}$**
95% CI for the difference: **[0.44, 0.93] g/dL**

**Interpretation:** Even though both groups show broadly comparable anemia prevalence (Task 3), their **mean HB levels are themselves significantly different** - elderly men average about 0.7 g/dL lower HB than young women. Since these two groups are compared against different WHO thresholds (13 g/dL for men vs. 12 g/dL for non-pregnant women), similar prevalence rates can mask different absolute severity. This is a subtle but important statistical nuance: **prevalence rate alone doesn't tell the full clinical severity story.**

**Organizational implications:** The client should be cautious about treating "similar prevalence" as "similar clinical urgency" across these two groups. Elderly male anemia may reflect distinct underlying causes (e.g. chronic disease, malignancy risk, nutritional deficiency in old age) that differ mechanistically from reproductive-age iron loss, and its lower absolute HB level plus smaller sample size (more uncertainty) suggests elderly male patients may warrant **separate clinical investigation protocols**, not just the same "iron deficiency" assumption applied to younger women.

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### 4.3 Comparison of Variances - Test C: HB Variance Across Institutions

**Hypotheses**

- $H_0$: Variance of HB is equal across all 4 institutions
- $H_1$: Variance of HB differs across at least one institution

**Test justification:** Comparing variances (not just means) across more than two groups calls for Levene's test or Bartlett's test. We report **both**: Levene's test (based on deviations from the group median) is robust to non-normality and is our primary test; Bartlett's test (based on the mean) is more powerful if normality holds, so we report it as a sensitivity check. This test also serves a second purpose: checking the **equal-variance assumption** required for the standard one-way ANOVA in Test F.

**Results**

| Institution | n | Variance (HB) | SD |
|---|---|---|---|
| **Institution_A** | 710 | 3.60 | 1.90 |
| **Institution_B** | 17,652 | 3.60 | 1.90 |
| **Institution_C** | 2,717 | 2.73 | 1.65 |
| **Institution_D** | 5,955 | 3.29 | 1.81 |

Levene's test: $\mathbf{F(3, 27030) = 32.63, p < 2.2 \times 10^{-16}}$
Bartlett's test: $\mathbf{K^2 = 90.82, df = 3, p < 2.2 \times 10^{-16}}$
**Interpretation:** Both tests agree - HB variance is **not equal** across institutions. Institution_C shows the smallest variance, meaning its HB readings are more tightly clustered, while Institutions A and B show the largest variance.

**Organizational implications:** This has two consequences. First, methodologically, it confirms that **Welch-type / variance-robust tests are the right choice** throughout this analysis rather than classical equal-variance tests. Second, from a quality-assurance perspective, we recommend the client investigate why HB variance differs systematically by institution - whether it reflects genuine population heterogeneity (e.g. Institution_C serving a more demographically narrow catchment) or measurement/equipment inconsistency across sites, since the latter would be a data-quality and clinical-standardization issue worth addressing operationally.

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### 4.4 Comparison of Proportions - Test D: Anemia Prevalence by Sex

**Hypotheses**

- $H_0$: Proportion anemic (Female) = Proportion anemic (Male)
- $H_1$: Proportion anemic (Female) ≠ Proportion anemic (Male)

**Test justification:** A two-proportion z-test (implemented via prop.test, chi-square-based with continuity correction) is the standard test for comparing a binary outcome rate between two independent groups - appropriate here since anemia status is binary (Y/N) and sex has two categories.

**Results**

| Sex | n | n Anemic | Proportion |
|---|---|---|---|
| **Female** | 15,597 | 8,455 | 54.2% |
| **Male** | 11,437 | 3,338 | 29.2% |

$\chi^2$ = 1,679, df = 1, $\mathbf{p < 2.2 \times 10^{-16}}$ 95%
CI for the difference (Female − Male): **[23.9%, 26.2%]**

**Interpretation:** Women in this dataset are anemic at roughly **1.85× the rate** of men (54.2% vs. 29.2%), a highly significant and large-magnitude difference - consistent with, and statistically confirming, the age-stratified pattern seen descriptively in Task 3.

**Organizational implications:** This is the single clearest, most quotable statistic for a Board-level presentation: **women are anemic at nearly double the rate of men** in this patient population. It provides strong, statistically defensible justification for prioritizing female-focused anemia screening and supplementation programs as a first-order resource allocation decision - while Test B/Task 3's age-interaction findings should prevent this from becoming the only focus (elderly men remain a high-risk group needing separate attention).

### 4.5 Comparison of Proportions - Test E: Anemia Prevalence Across Institutions

**Hypotheses**

- $H_0$: Anemia status is independent of institution
- $H_1$: Anemia status is **not** independent of institution

**Test justification:** With more than two groups (4 institutions) and a binary outcome, a **chi-square test of independence** on the institution × anemia contingency table is the appropriate test (rather than repeated pairwise proportion tests, which would inflate Type I error without correction).

**Results**

| Institution | % No | % Yes | n |
|---|---|---|---|
| **Institution_A** | 54.5% | 45.5% | 710 |
| **Institution_B** | 57.9% | 42.1% | 17,652 |
| **Institution_C** | 42.7% | 57.3% | 2,717 |
| **Institution_D** | 58.4% | 41.6% | 5,955 |

$\chi^2 = 233.85$, df = 3, $\mathbf{p < 2.2 \times 10^{-16}}$
**Cramer's V = 0.093** (small-to-moderate effect size)

Standardized residuals show **Institution_C** as the dominant driver of the association (residual ≈ +15.2 for "Yes"/anemic cell - far exceeding the ±2 threshold typically used to flag a cell's contribution), with Institution_B and D showing smaller, opposite-direction contributions.

**Interpretation:** The association between institution and anemia status is statistically significant, but the **effect size is small** (Cramer's V = 0.093, conventionally "small" on Cohen's scale) - meaning while institution is not independent of anemia status, it is a comparatively weak predictor on its own compared to sex or age. The practical driver is almost entirely Institution_C's unusually high prevalence (57.3% vs. ~42-46% elsewhere).

**Organizational implications:** This finding - statistically real but small in magnitude and concentrated in one institution - is exactly the kind of nuance a client needs spelled out rather than a bare "p < 0.05, therefore institutions differ" statement. We recommend the client investigate Institution_C specifically (patient demographics,

catchment area, referral patterns) rather than assume systemic differences across all institutions.

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### 4.6 One-Way ANOVA - Test F: HB Across Age Groups

**Hypotheses**

- $H_0$: Mean HB is equal across all 8 age groups
- $H_1$: At least one age group's mean HB differs from the others

**Test justification:** With more than two independent groups (8 age bands) and a continuous outcome (HB), one-way ANOVA is the appropriate test. Task 3 confirmed HB is approximately symmetric within groups (supporting the normality assumption reasonably well at this sample size), and while Test C shows variances differ somewhat by subgroup, ANOVA is known to be robust to moderate variance heterogeneity at these large, only mildly unequal sample sizes. We supplement the test with **Tukey's HSD post-hoc test** to identify exactly which age groups differ, since a significant ANOVA alone does not indicate where the differences lie.

**Results**

| Age group | n | Mean HB | SD |
|---|---|---|---|
| **0–4 (Child, U5)** | 2,120 | 11.16 | 1.21 |
| **5–11 (Child)** | 2,004 | 12.06 | 1.18 |
| **12–14 (Adolescent)** | 479 | 12.53 | 1.27 |
| **15–17 (Adolescent)** | 671 | 12.86 | 1.77 |
| **18–39 (Young adult)** | 9,347 | 12.94 | 1.88 |
| **40–59 (Middle-aged)** | 6,707 | 12.52 | 1.91 |
| **60–79 (Older adult)** | 5,048 | 11.78 | 1.80 |
| **80+ (Elderly)** | 658 | 11.03 | 1.77 |

**$F(7, 27026) = 398.1$, $p < 2.2 \times 10^{-16}$**
**Eta-squared = 0.094** (age group explains ~9.4% of HB variance - a moderate effect for a single demographic factor)

Tukey HSD (selected pairwise results, all family-wise-corrected):

- Most pairwise comparisons are significant ($p < 0.05$ to $p < 0.001$).
- Notable **non-significant** pairs: 18–39 vs. 15–17 ($p = 0.96$), 40–59 vs. 12–14 ($p = 0.52$) - these adjacent groups do not differ reliably.
- The **largest gaps** are between young adults (18–39, peak HB) and both extremes: 80+ elderly (diff ≈ −1.86 g/dL) and 0–4 children (diff ≈ +1.62 g/dL in the other direction).

Shapiro-Wilk test on a 5,000-row residual sample: $W = 0.998$, $p = 2.7 \times 10^{-6}$ - technically significant (departure from perfect normality), but with this sample size even trivial deviations reach significance; visual inspection (Task 3 histograms) shows the departure is minor and ANOVA remains a reasonable, standard choice at this scale.

**Interpretation:** HB follows a clear **age trajectory**: rising through childhood and adolescence, peaking in young adulthood (18–39), then declining steadily through middle and older age, reaching its lowest point in the 80+ group. The ANOVA confirms it is not due to chance.

**Organizational implications:** Age-based HB norms are not linear or binary - a "young vs. old" split is too coarse. This result supports building **age-band-specific reference expectations** into any screening tool rather than treating age as a simple linear predictor and gives the client concrete evidence for age-stratified resource and supplementation planning - e.g., recognizing both **young children** and the **very elderly** as physiologically distinct lower-HB risk groups requiring different interventions (nutritional vs. geriatric/chronic-disease-focused).

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**Summary Table: All Tests**

| Test | Type | Result | Practical takeaway |
|---|---|---|---|
| **A** | Means (HB, Sex) | p < 2.2e-16 | Men have ~1.65 g/dL higher HB; sex-specific thresholds justified |
| **B** | Means (young F vs elderly M) | p = 1.09e-7 | Similar prevalence masks different absolute HB severity |
| **C** | Variances (HB, Institution) | p < 2.2e-16 | Variance differs by institution; supports robust-test choice; QA flag for the client |
| **D** | Proportions (Anemia, Sex) | p < 2.2e-16 | Women anemic at ~1.85× the rate of men |
| **E** | Proportions (Anemia, Institution) | p < 2.2e-16, small effect (V=0.093) | Real but weak association, driven mainly by Institution_C |
| **F** | One-way ANOVA (HB, Age group) | p < 2.2e-16, $\eta^2$=0.094 | Clear age trajectory: peaks in young adulthood, declines at both extremes |